Candidate Intelligent Ranking System

India Runs Data & AI Challenge — Redrob Hackathon 2026

An AI-powered multi-dimensional candidate ranking engine that evaluates candidates the way a great recruiter would — holistically, not just by keywords.

The Problem with Keyword-Based Hiring

■ **Keyword Blindness**	ATS systems filter 75% of qualified resumes before a human sees them. Great candidates with non-standard phrasing get dropped — not because they can't do the job, but because they wrote 'ML' instead of 'Machine Learning'.
■ **Uni-Dimensional Scoring**	Most systems score candidates on skill keywords alone, ignoring career trajectory, platform signals, cultural contribution (GitHub, LinkedIn), and practical fit signals like notice period and salary expectations.
■■ **Recruiter Overload**	A recruiter reviewing 500+ profiles for a single role spends ~6 seconds per profile. The result: qualified candidates are missed, and the hiring cycle extends by weeks.
■ **The Hidden Signal Gap**	A candidate's GitHub contributions, open-source impact, and professional network are rich behavioral signals that keyword systems completely ignore.

CIRS — Our Solution

CIRS scores every candidate across **5 weighted dimensions**, each capturing a different signal of real-world job fit. The result: a recruiter-trusted shortlist in seconds, not hours.

■ **Skill Match**	**35%**	Semantic + exact coverage of JD requirements. Required skills weighted 2x over nice-to-have.
■ **Experience**	**25%**	YOE alignment, company prestige tier, career progression & tenure analysis.
■ **Platform Signals**	**20%**	GitHub (stars, contributions, PRs) + LinkedIn (recommendations, activity, network).
■ **Education**	**10%**	College tier (IIT/NIT/BITS/Tier-2/3), CGPA score, certifications earned.
■ **Practical Fit**	**10%**	CTC range alignment, notice period, location match.

System Architecture

INPUT	PROCESSING	OUTPUT
Job Description (JSON)	Skill Match Scorer (synonym normalization)	Composite Score (0-100)
500+ Candidate Profiles (JSON/CSV)	Experience Scorer (YOE + prestige tiers)	Ranked CSV (all candidates)
	Platform Signal Scorer (log-scale GitHub/LI)	Top-50 Shortlist
	Education Scorer (tier + CGPA)	Summary JSON (top-10 + stats)
	Fit Scorer (CTC, notice, location)	

Tech Stack

Python 3.11+	Pandas / NumPy	Scikit-learn	JSON / CSV I/O

Scoring Deep Dive

Skill Match (35%)

Required skills are matched using synonym normalization: 'PyTorch' = 'torch' = 'pytorch'. Required skills contribute 70% to skill score; nice-to-have contribute 30%. This prevents keyword-stuffed resumes from gaming the system.

Experience (25%)

YOE is scored against the JD range with asymmetric penalties — over-qualification is penalized lightly (senior fits junior roles somewhat); under-qualification is penalized more steeply. Company prestige uses 5 tiers: FAANG > Unicorn > Growth > Enterprise > Startup. Most recent employer weighted 1.5x.

Platform Signals (20%)

GitHub: star count uses log-scale normalization (1k stars vs 100k stars are very different signals). Contributions and merged PRs reflect active open-source engagement. LinkedIn: recommendations are weighted most (peer validation), followed by connection count and endorsements.

Education (10%)

Tiers: IIT (100) > BITS (85) > NIT (75) > Tier-2 (55) > Tier-3 (35). CGPA scaled linearly from 5.0 (min useful) to 9.8 (max). Each certification adds up to 25 points (capped at 100).

Practical Fit (10%)

CTC below budget is a mild positive; above budget penalized 8 pts/lpa over range. Notice period <= max_days is full marks; every day over is -2 pts. Bangalore candidates for Bangalore roles score 100; remote-ok scores 85 for hybrid.

Results on Sample Dataset

500	63.6	51.7	39.9
Candidates Ranked	Top Score	Mean Score	Lowest Score

Sample Top-10 Rankings

Rank	Candidate ID	Composite	Skill Match	Experience	Platform	Stage
1	CAND_0281	**63.6**	38.1	77.4	85.5	mid
2	CAND_0186	**63.5**	37.9	91.5	84.0	mid
3	CAND_0438	**62.3**	36.1	93.0	59.9	mid
4	CAND_0111	**62.2**	40.7	79.0	60.0	mid
5	CAND_0126	**62.1**	47.5	95.8	47.1	senior

Key Design Decisions

Q: Why multi-dimensional scoring?

A: No single signal predicts job success. A great GitHub profile with weak skills is different from weak GitHub with perfect skill match. Weighted combination reflects how experienced recruiters actually think.

Q: Why synonym normalization?

A: Real resumes are inconsistent. 'PyTorch', 'pytorch', 'torch' all mean the same thing. Without normalization, valid candidates get penalized for formatting, not ability.

Q: Why log-scale for GitHub stars?

A: The difference between 0 and 100 stars is huge; the difference between 5,000 and 10,000 is marginal. Log normalization prevents superstar outliers from overwhelming the signal.

Q: Why penalize over-qualification lightly?

A: A senior engineer applying to a mid-level role may be relocating, changing industries, or prefer a lower-pressure environment. Automatic rejection loses great candidates on a technicality.

Q: Why weight recent employers 1.5x?

A: A candidate who spent 3 years at TCS and just joined Google is different from a 10-year TCS lifer. Recency-weighting captures trajectory, not just history.

Roadmap — What We'd Build Next

v1.1 — Semantic Skill Matching	Replace synonym dictionaries with sentence-transformers embeddings. Compute cosine similarity between JD skill descriptions and candidate skill vectors. Catches 'built NLP pipelines' even without explicit skill labels.
v1.2 — LLM-Powered Fit Reasoning	Send top-50 shortlisted profiles + JD to Claude/GPT for narrative fit assessment. Generate a 2-sentence recruiter note per candidate explaining WHY they ranked where they did.
v1.3 — Behavioral Signal Parsing	Parse cover letters, portfolio bios, and GitHub README files for motivation signals: passion for the domain, communication quality, growth mindset indicators.
v2.0 — Feedback Loop Training	Record recruiter accept/reject decisions on shortlisted candidates. Fine-tune dimension weights using logistic regression on historical hiring outcomes. Self-improving system that gets better with every hiring cycle.

Make Hiring Smarter

- 500 candidates scored across 5 dimensions in under 2 seconds
- Recruiter-ready shortlist: Top 50 with full score breakdown
- Explainable scores: every candidate gets dimension-level transparency
- No black-box: weights are visible, tunable, and defensible
- Built to extend: embeddings, LLM reasoning, and feedback loops ready to plug in

Submission: GitHub Repo | ranked_candidates.csv | shortlist_top50.csv
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